

Problem-Solution Fit

Date	1 July 2026
Team ID	
Project Name	Voice-Based Communication Understanding Analyzer
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

 PROBLEM – SOLUTION FIT CANVAS Voice Based Communication Understanding Analyzer 		
1. CUSTOMER SEGMENT(S) [CS] - Students (School, College, University) - Educators / Teachers / Trainers - Interview Candidates / Job Seekers - Professionals / Corporate Employees - Educational Institutions - Researchers in NLP, Education & AI	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited budget for paid tools or subscriptions - Basic devices (laptops, mobiles) with average performance - Internet dependency for some AI models - Limited technical knowledge - Time constraints for practice and evaluation - Preference for easy-to-use solutions	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions: - Manual evaluation by teachers - Speech-to-text tools (e.g., Google STT) - Online quiz / test platforms - Presentation evaluation tools Pros: Easy to use, widely available, familiar tools Cons: No semantic understanding evaluation, no speech fluency analysis, subjective results, limited personalized feedback
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty in expressing and explaining concepts verbally - No objective evaluation of conceptual understanding - Lack of feedback on speech fluency, clarity and coherence - Subjective and inconsistent evaluations - Delayed feedback from instructors or peers - Occurs frequently during exam preparation, presentations, interviews and self-study	9. PROBLEM ROOT / CAUSE [RC] Traditional assessment methods mainly focus on written answers and ignore spoken communication. There is a lack of automated systems that can analyze speech, understand the meaning (semantics) of spoken content, evaluate fluency and provide instant, objective feedback to improve conceptual understanding and communication skills.	7. BEHAVIOR + ITS INTENSITY [BE] - Students regularly practice speaking concepts before exams and interviews - Seek validation and feedback on their understanding - High intensity during exam/interview seasons - Need instant, reliable and unbiased evaluation - Regular usage for self-improvement and confidence building
3. TRIGGERS TO ACT [TR] - Upcoming exams and assessments - Interview and placement preparation - Classroom presentations and viva - Concept revision and self-assessment - Desire to improve communication skills - Need for objective, instant feedback	10. YOUR SOLUTION [SL] Voice Based Communication Understanding Analyzer An AI-powered system that: - Converts speech to text using Whisper - Evaluates conceptual understanding using Sentence-BERT - Analyzes speech fluency using Librosa - Generates comprehension score and qualitative feedback - Visualizes waveform for better insights - Generates downloadable PDF reports for students and educators	8. CHANNELS OF BEHAVIOR [CH] ONLINE - Web Application (Streamlit) - College LMS / Portals - Educational Websites - Browser Access OFFLINE - Classroom Presentations - Laboratory Sessions - Workshops / Seminars - Interview Practice Sessions
4. EMOTIONS BEFORE / AFTER [EM] BEFORE:  Nervous  Confused  Low Confidence  Uncertain about Understanding  Anxious before Presentations / Interviews AFTER:  Confident  Motivated  Clear Understanding of Concepts  Improved Communication Skills  Ready for Exams / Interviews		